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Clockwise and anticlockwise rotation of Viti Levu, Fiji—in relation to the tectonic development of the North and the South Fiji Basin

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SUMMARY

Samples for palaeomagnetic study were collected from 40 sites in Viti Levu, Fiji. Reliable palaeomagnetic directions were obtained from 22 sites after magnetic cleaning. These data cover the age between late Eocene to Pliocene. Geological study indicates that Viti Levu can be structurally divided into three stages; stage 1 (late Eocene to early Oligocene), stage 2 (early to middle Miocene) and stage 3 (late Miocene to Pliocene). The mean palaeomagnetic direction (D, I) for each stage is (−10°, −48°) in stage 1, (−64°, −28°) in stage 2 and (−32°, −45°) in stage 3, respectively.

Expected directions of the geomagnetic field in the past were calculated from the Australian apparent polar wander path (APWP) and were subtracted from the obtained palaeomagnetic directions in geological sequence. Then, the tectonic movement in and around the island of Viti Levu with respect to the Australian Plate was estimated for post-Eocene time. The following conclusions were reached. (1) A clockwise rotation of about 45° took place during Oligocene, (2) an anticlockwise rotation of about 75° had occurred since the end of the middle Miocene time, and (3) the Euler poles of these rotations were situated in or near Fiji.

A tectonic history of the Fiji region since late Eocene, which can explain the episodes of clockwise and anticlockwise rotations of Viti Levu, is as follows: clockwise rotational movement in Oligocene was closely related to the formation of the South Fiji Basin and anticlockwise rotational movement since the end of middle Miocene was closely related to the formation of the North Fiji Basin. The clockwise rotation of Viti Levu was accompanied by fan-shaped spreading in the northern half of the South Fiji Basin, in which a system, ridge–ridge–ridge (R–R–R) triple junction of plates, was involved. The subsequent anticlockwise rotation of Viti Levu can be explained as 'ball-bearing' rotation associated by antithetical transform faults accompanied with the spreading of the North Fiji Basin.

Key words: North Fiji Basin, palaeomagnetism, rotation, South Fiji Basin, Viti Levu.

1 INTRODUCTION

Fiji is one of the island nations located in the southwest Pacific. The largest island, Viti Levu, and the second largest island, Vanua Levu, are encircled by about 320 smaller islands. Several types of topographic units border and subdivide the area surrounding Fiji (Fig. 1), suggesting that the tectonic history of this area is not simple. Within a region between the New Hebrides and the Tonga–Kermadec trench-island arc systems (simply 'the Fiji region'), there are three basins, i.e. the North Fiji Basin, the South Fiji Basin and the Lau Basin–Havre Trough. The North Fiji Basin and the South Fiji Basin are divided by the Hunter Fracture Zone and the South Fiji Basin and the Lau Basin–Havre Trough are divided by the Lau–Colville Ridges. The Fiji islands are on the junction of the Hunter Fracture Zone and the Lau Ridge. Explanations of the

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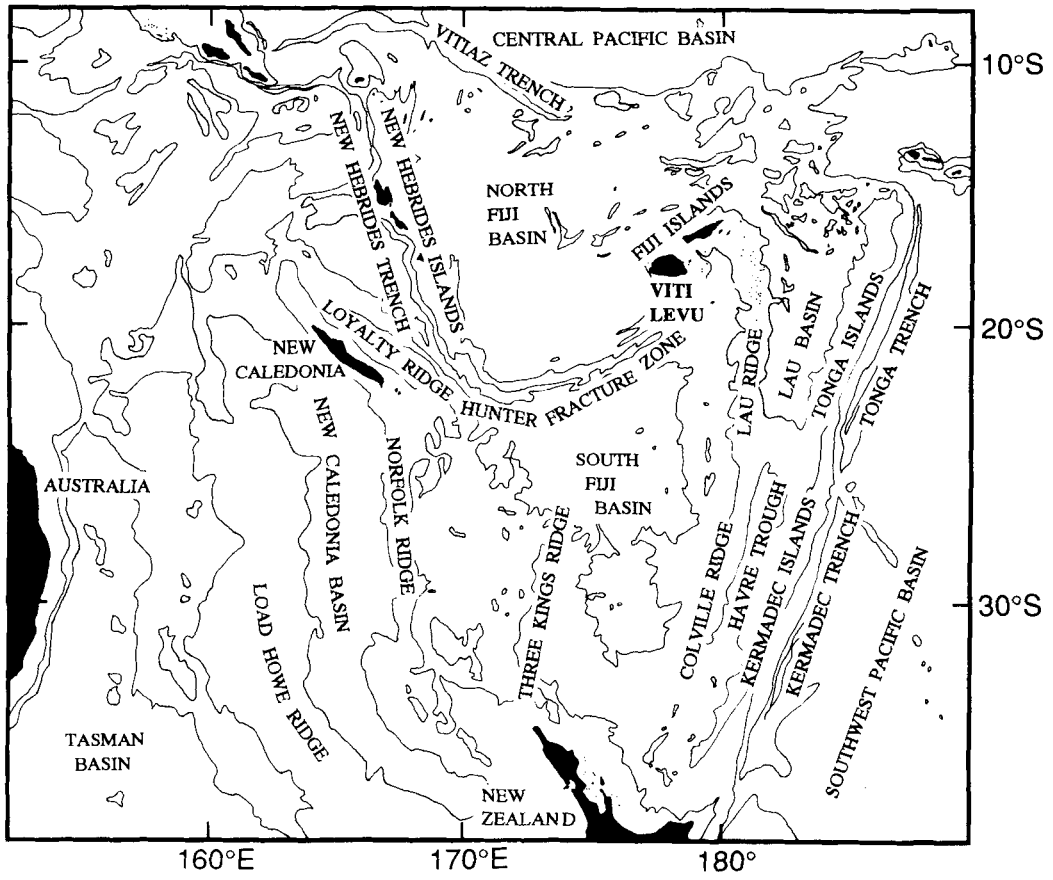


Figure 1. Bathymetric map of the adjacent sea of Viti Levu, Fiji. Water depth is shown by the 2000 m contour. Data for the map are taken from the International Hydrographic Organization (1982).

origin and development of the complex physiographic and geologic structure of the region have been made within the framework of the theory of plate tectonics (e.g. Chase 1971; Green & Cullen 1973; Kroenke 1984; Brocher 1985a). It may be concluded from their discussion that: (1) two major plates, the Australian and the Pacific Plates, have been converging in the Fiji region since Palaeogene, and (2) the plate boundaries in the convergence region were repeatedly reorganized accompanied by various tectonic events, e.g., back-arc spreading, roll-back of trenches and initiation, termination of trenches and so on. The tectonic setting of Fiji islands has repeatedly changed by such reorganizations of the plate boundaries. Therefore, it is important to unravel the tectonic history of Viti Levu to understand the tectonic history of the Fiji region. The tectonics of the region may be a clue to understanding a number of fundamental problems (Uyeda 1986) in the subduction tectonics.

A paleomagnetic study of Viti Levu is particularly appropriate to unravel the tectonic history of the island region, because palaeomagnetic methods are generally useful to detect and measure relative movement and rotation of different tectonic units on a global and a subcontinental scale. Viti Levu is the most suitable island for palaeomagnetic study in the Fiji archipelago, because the oldest rocks (late Eocene) of the Fiji archipelago crop out in Viti Levu and various volcanic and sedimentary rocks, ranging from late Eocene to Pleistocene, are widely distributed in the island (Rodda 1967). The remanent magnetization of rocks from Viti Levu, Fiji has been described by several authors (Tarling 1967; James & Falvey 1978; Malahoff *et al.* 1982b). Malahoff (1970) made a speculative proposal that the Fiji Platform had undergone anticlockwise rotation based on earthquake distribution and submarine morphology, and some of the palaeomagnetic results seem to support this hypothesis (James & Falvey 1978; Malahoff *et al.* 1982b).

The present paper presents further results from some sites close to those of previous authors', and also from different sites. The purpose of the present study is to investigate more thoroughly the direction of magnetization of rocks distributed in Viti Levu from late Eocene to Pliocene, and to infer tectonic movements in and around Viti Levu.

2 OUTLINE OF GEOLOGY

The geology of Fiji has been summarized by Rodda (1967, 1974, 1975) and, more recently, by Rodda & Kroenke (1984) and Rodda (1992), with an important contribution by Hathway (1992). From these, we summarize the geology as follows.

The oldest rocks in Fiji are found in western Viti Levu, and are classified as the Yavuna Group. They are basaltic to basin-andesitic massive and pillowed lavas, pillow breccias and reworked volcaniclastic rudites associated with limestones containing faunas indicating both Tertiary *b* (late

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Eocene) and *c* (early Oligocene) stages. Some of the Eocene lavas are geochemically primitives, but more-evolved (arc-tholeiitic) types occur, and the lavas associated with the Oligocene limestones have a similar evolved geochemistry. Minor decitic rocks also occur. The group includes the Yavuna Stock, of tonalite (34 Ma) intruded by a swarm of dolerite dykes which are probably of similar age. Late Eocene rocks also possibly occur in southern Viti Levu.

The oldest well-dated rocks of the succeeding Wainimala Group are late Oligocene to early Micoene; these lie with angular unconformity on Yavuna Group rocks. An exposure of volcaniclastic turbidites which appear to rest unconformably on Yavuna rocks has a nannoflora indicating zone CN18, late early Oligocene, so the lacuna between the Yavuna and Wainimala rocks might be restricted to part of the early Oligocene rather than including part of the late Oliogcene as well. The Wainimala Group rocks, largely basaltic (including pillow lavas) and dacitic, are island-arc tholeiitic in nature and form most of the deformed 'basement' of the island. The youngest rocks of the group are middle Miocene in age.

The Savuna Volcanic Group, in the southwest, consists of basalt to dacite, similar in geochemistry to Wainimala Group rocks. It is thought to be no younger than middle Miocene but there is no direct evidence for age, the only constraint being some rocks of the group are overlain by strata considered to be late Miocene in age, and others are overlain by definite earliest Pliocene strata. Yaskawa *et al.* (1985) reported a data of 6.5 ± 0.5 Ma for rock throughout belonging to the Savuna Volcanic Group; though this is rather younger than the presumed age, it is not in conflict with the local stratigraphy. Although most contacts of Savuna rocks with Wainimala rocks are faulted, an angular unconformity exists in places.

Gabbro and tonatile stocks of the Colo Plutonic Suite crystallized in the middle and late Miocene (known range is 12.5 to about 7.5 Ma). They were emplaced synorogenically, mainly in anticlinal folds, in the Wainimala Group across southern and central Viti Levu; widespread metamorphism up to greenschist faces occurred at the same time.

Rocks of the Wainimala Group are overlain disconformably, with no apparent discordance, by sandstones and conglomerates assigned to the Tuva Group. These rocks are of late Miocene (CN9) age and are relatively highly deformed in places.

The rocks of these groups are unconformably overlain by relatively undeformed sedimentary rocks with some andesite, of the Ra, Nadi and Navosa Sedimentary Groups, the Medrausucu Group and the Vatukoro Greywacke (Ba Volcanic Group). Calke-alkaline andesites were erupted in southeastern Viti Levu between 6 and 5.5 Ma, and high-K andesites were erupted further west at about the same time.

A belt of younger eruptive and intrusive rocks ranging in age from 5.5 to 3 Ma trends E–NE across nothern Viti Levu. The belt contains the Koroimavua and Ba Volcanic Groups and includes associated monzonite and olivine monzonite stocks and sills. The Koroimavua Volcanic Group is latest Miocene to earliest Pliocene and overlies the Nadi Group. The Ba Volcanic Group is mostly basalt with some andesite and differentiates, and originated from several volcanic centres in the north. The two largest volcanoes are the shoshonitic Tavua volcano in central north and the calc-alkaline Rakiraki volcano in the northeast. The rocks of the group are mostly early Pliocene, but some of the basal rocks may be late Miocene.

The Verata Sedimentary Group in the east and the Cuvu Sedimentary Group in the southwest are both early Pliocene with late Pliocene and probably some Pleistocene strata in the east of Viti Levu.

3 PALAEOMAGNETISM

3.1 Sample collection

Orientated rock samples for palaeomagnetic study were collected in Viti Levu during 1982 (19 sites) and 1985 (21 sites). Their approximate locations are shown in Fig. 2 and the sampling details are tabulated in Table 1.

The samples were collected with the use of a magnetic compass for orientation. In the laboratory, they were drilled and sliced into cylindrical specimens 2.5 cm both in diameter and in length. A geomagnetic declination of 13.0° (1982) and 13.2° (1985), inferred for the island of Viti Levu from the IGRF (IAGA Division I Working Group 1 1985), was added to obtain the strike direction from geographic north.

The orientation of the bedding plane of the rock-body, from which palaeomagnetic samples were collected, was measured at each site as follows. In the case of layered sedimentary rocks, the bedding orientation was determined as the mean of orientation of several layers. For massive rocks, it was estimated from the bedding orientation of structurally associated strata. In some cases, it was not possible to determine the original attitudes of rocks.

3.2 Measurements and results

The direction and intensity of natural remanent magnetization (NRM) was measured with a cryogenic magnetometer. Some comparatively strongly magnetized specimens of some volcanic rocks were measured with a spinner magnetometer.

All the specimens were demagnetized in an alternating field in the following way. A few specimens from each site were progressively demagnetized in steps until a stable direction of magnetization was reached (see Fig. 3). The remaining specimens from the same site were then demagnetized in the field which had thus been determined. For those sites where the direction of magnetization in the specimens continued to change on application of higher fields, several more specimens were progressively demagnetized to determine the peak alternating field at which the dispersion was smallest.

For these sites where the dispersion of magnetic vectors was not improved through the above-mentioned AF demagnetization procedure, thermal demagnetization was attempted. That is, the specimens were progressively heated up to the temperature at which their intensity of magnetization was reduced to 5 per cent of the original one or the dispersion of magnetic vectors was smallest. In some cases the thermal demagnetization was more effective than the AF demagnetization to isolate the stable direction. Typical examples of magnetic behaviour upon stepwise AF and thermal demagnetization are shown in Fig. 3 using orthogonal plots.

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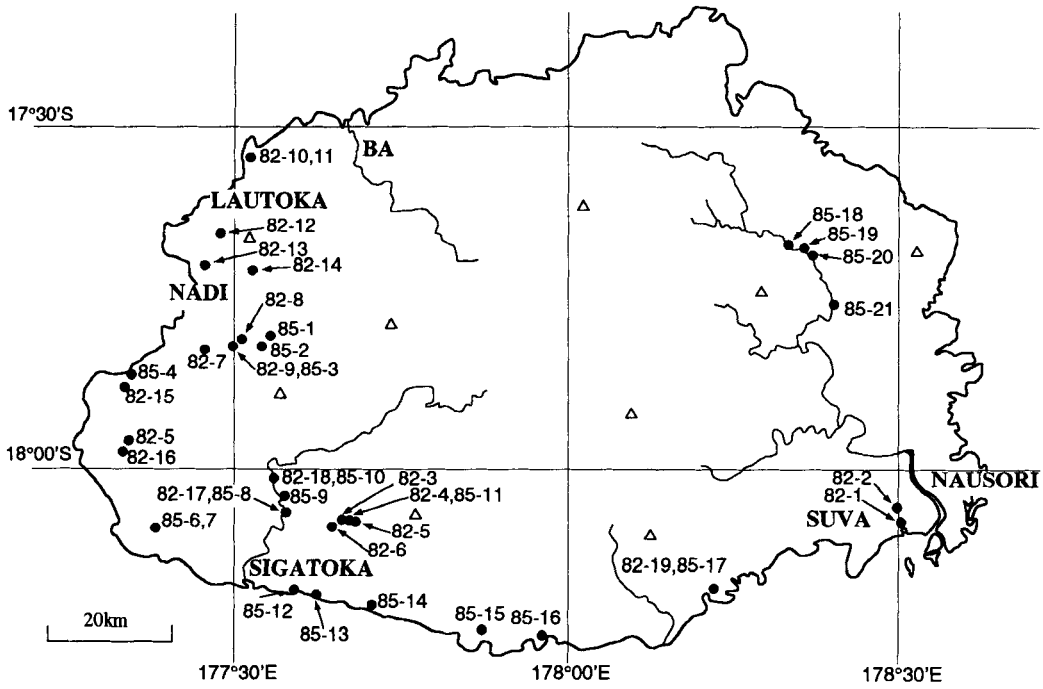


Figure 2. Map of Viti Levu, Fiji, showing locations of palaeomagnetic sampling sites. Numerals are site numbers.

The mean magnetic directions, intensities and statistical parameters are presented in Table 2. After completion of demagnetization treatment the volume susceptibility κ of each specimen was determined from magnetization induced by a known field trapped inside the cryomagnetometer. The index of the specimen's capability of maintaining a stable remanence, the Koenigsberger ratio Qr was calculated as $Qr = NRM/\kappa H$, where H is the geomagnetic field at each sampling site, inferred from IGRF (IAGA Division I Working Group 1 1985). The mean Koenigsberger ratio and the mean median destructive field (MDF) for all the specimens at each sampling site are shown in Table 2.

3.3 Reliability of results

The magnetic and microscopic properties of the samples do not show any particular characteristics, indicating either remarkable stability of magnetization or useless instability (see Table 2). Stable magnetization was observed in some sedimentary rocks and, on the contrary, unstable magnetization was observed in some volcanic rocks. In many cases, instability of NRM may be due to weathering as suggested by Tarling (1967).

The Koenigsberger ratio Qr usually ranges between 2 to 20 in volcanic rocks and >3 in sedimentary rocks. The values of MDF (Table 2) are rather low on the whole, but some of these low MDF samples show good grouping of the direction of their remanent magnetization after AF demagnetization. This indicates the presence of some magnetically unstable components, i.e. an isothermal or viscous remanent magnetization, which can be removed by applying an adequate alternating field.

Microscopic studies show that all samples contain a magnetic spinel phase mineral, presumably titanomagnetite, but almost no rhombohedral or hexagonal minerals. The peripheral regions of some titanomagnetite may have been altered into titanomaghemite by low temperature oxidation, as demonstrated by colour variation.

The grain size of magnetic minerals ranges between 2 and 250 μm as shown in Table 2 and does not seem to show any correlation with other stability indices. Since the indices of magnetic stability have no clear correlation with each other, it may be the best way to estimate the reliability of the direction of magnetization with the value of α_{95} (Fisher 1953).

3.4 Characteristic palaeomagnetic directions

Reliable palaeomagnetic directions were obtained from 22 sites, on the bases of five acceptance criteria.

(1) Each site has to consist of at least three individually oriented samples, from which a few specimens were measured.

(2) The value of α_{95} has to be smaller than 20° for each site.

(3) The mean direction of magnetization for each site must not coincide with the present geomagnetic field before bedding correction.

(4) The mean direction of magnetization for each site has to coincide with the direction of magnetization obtained from the progressive demagnetization data of a few pilot specimens. The direction of magnetization is determined from a straight line that passes through three or more data points and the origin on an orthogonal vector plot.

(5) The mean direction of magnetization in each site has not to be a record of a transition of geomagnetic field (site 85–15).

A mean direction for time intervals was determined for reliable site-mean directions of corresponding age after

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Table 1. Review of sampling sites.

site	N	rock kind	name of group	age	remarks
82-1	3	siltstone	Suva Marl	Pliocene	Rodda et al., 1985
82-2	5	basalt	Savura V. G.	latest Oligocene to early middle-Miocene	geological estimation
82-3	7	limestone	Wainimala G.	early middle-Miocene	Coleman, 1974
82-4	6	basalt	Wainimala G.	early middle-Miocene	geological estimation
82-5	5	basalt	Wainimala G.	early middle-Miocene	geological estimation
82-6	6	volcanics	Wainimala G.	early middle-Miocene	geological estimation
82-7	6	sandstone	Nadi S. G.	late Miocene	NN.11 by Wilcoxon (Easton, 1973)
82-8	5	limestone	Yavuna G.	late Eocene	Cole, 1960
82-9	6	volcanics	Yavuna G.	late Eocene	geological estimation
82-10	4	basalt	Ba V. G.	4.73 Ma	Whelan et al., 1985
82-11	5	basalt	Ba V. G.	4.73 Ma	Whelan et al., 1985
82-12	7	basalt	Ba V. G.	5.04 Ma	Yaskawa et al., 1985
82-13	7	volcanics	Koroimavua V. G.	5.4 Ma	eruptive rocks nearby dated at 5.4 Ma (McDougall, 1963)
82-14	7	micromonzonite	Koroimavua V. G.	4.9 Ma	Rodda & Kroenke, 1984
82-15	5	basalt	Wainimala G.	late Oligocene to early middle-Miocene	geological estimation
82-16	7	gabbro	Colo P. Suite	middle Miocene	geological estimation
82-17	5	dacite	Wainimala G.	middle Miocene	geological estimation
82-18	5	andesite	Wainimala G.	middle Miocene	geological estimation
82-19	1	andesite	Medrausucu G.	5.85 Ma	Gill & McDougall, 1973
85-1	5	mudstone	Yavuna G.	Eocene	geological estimation
85-2	6	dolerite	Yavuna G.	34 Ma	McDougall, 1963
85-3	5	sedimentary	Yavuna G.	Eocene	geological estimation
85-4	5	sandstone	Wainimala G.	late Oligocene to early Miocene	Hathway, in press
85-5	5	gabbro	Colo P. Suite	middle Miocene	geological estimation
85-6	5	volcanics		middle Miocene	geological estimation
85-7	5	mudstone		middle Miocene	geological estimation
85-8	5	dacite	Wainimala G.	middle Miocene	geological estimation
85-9	5	sandstone	Wainimala G.	early to middle Miocene	geological estimation
85-10	6	andesite	Wainimala G.	early to middle Miocene	geological estimation
85-11	6	basalt	Wainimala G.	early middle-Miocene	geological estimation
85-12	5	sandstone	Wainimala G.	middle Miocene	geological estimation
85-13	5	sandstone	Wainimala G.	middle Miocene	geological estimation
85-14	5	andesite	Wainimala G.	early to middle Miocene	geological estimation
85-15	6	tuff	Wainimala G.	Eocene	geological estimation
85-16	6	tuff	Wainimala G.	Eocene	geological estimation
85-17	8	andesite	Medrausucu G.	5.85 Ma	Gill & Mcdougall 1973
85-18	5	sandstone	Wainimala G.	early middle-Miocene	paleontology
85-19	6	sandstone	Wainimala G.	early middle-Miocene	paleontology
85-20	5	mudstone	Wainimala G.	early middle-Miocene	paleontology
85-21	5	limestone	Wainimala G.	Eocene	paleontology

S. : Sedimentary, V. : Volcanic, G. : Group, P. : Plutonic

correction for bedding where possible. Reversed polarity directions were inverted to their normal polarity equivalent. Polarities were judged from the sign of the inclination. A positive inclination represents a reversed polarity, as Fiji has resided throughout the Tertiary in the southern hemisphere.

Two major unconformities, one Oligocene and one late Miocene, are recognized in Viti Levu. Using these unconformities all palaeomagnetic data were divided into three time stages, based on radiometric and geologic ages of the rock samples. Stage 1 ranged from late Eocene to early Oligocene, stage 2 ranged from late Oligocene to middle Miocene, and stage 3 ranged from late Miocene to Pliocene. The mean palaeomagnetic direction for each stage is tabulated in Table 3 and shown in Fig. 4.

3.4.1 Stage 1 (late Eocene to early Oligocene)

The rocks from five sites of the Yavuna Group including a dyke in the tonalite stock have preserved reliable primary palaeomagnetic direction. The declination of the mean palaeomagnetic direction is deflected 10° west with respect to the present geocentric dipole field ($D = 0.0°$, $I = -33.0°$). Although the geocentric dipole field direction lies within a 95 per cent confidence limit of the stage mean, the primary value of this palaeomagnetic direction is confirmed by the presence of both normal and reversed polarities, by the good agreement between both palaeomagnetic directions from igneous rocks and from sedimentary rocks. The reliability of the mean direction is also ascertained through the evidence that the directions before tectonic correction from all the sites in this stage are different from the present geomagnetic field.

3.4.2 Stage 2 (late Oligocene to middle Miocene)

The declination of the mean direction ($D = -63.7°$, $I = -28.3°$) from 12 reliable sites is clearly deflected to the west, and the inclination is similar to that of the present geocentric axial dipole field. The 12 reliable palaeomagnetic directions were obtained mainly from both volcanic and sedimentary rocks of the Wainimala Group and the Savuna Volcanic Group, and also from some stocks of the Colo Plutonic Suite.

The site-mean directions are grouped into two separate

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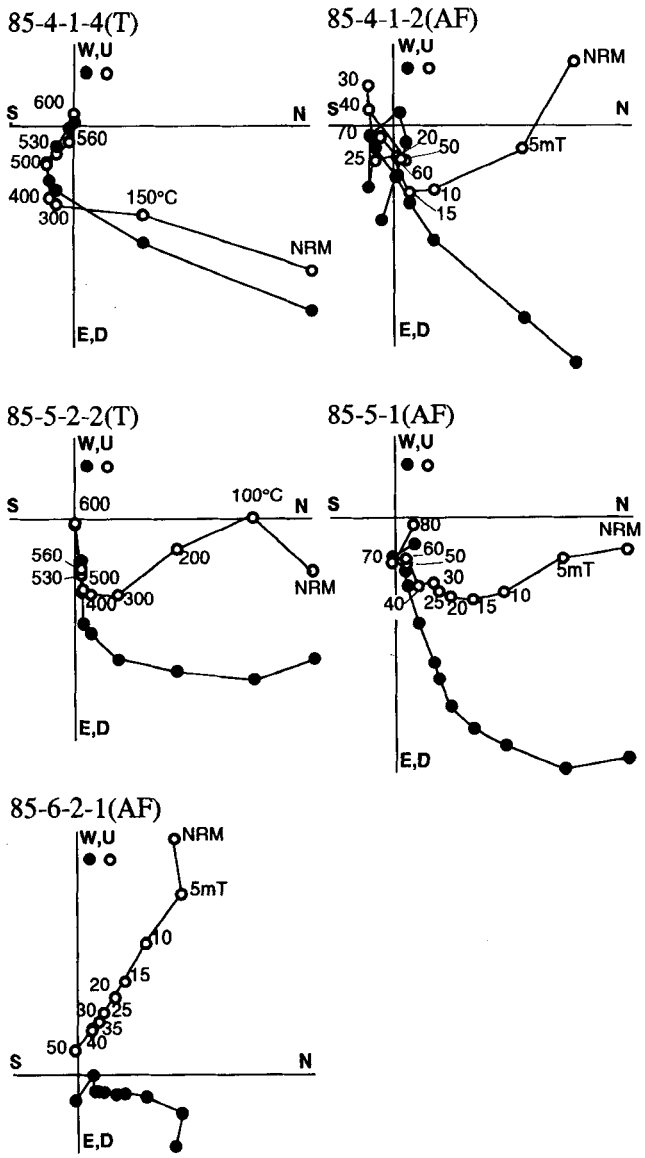


Figure 3. Typical examples of the directional change of remanent magnetization projected onto two orthogonal planes (Zijderveld 1967) corresponding to progressive thermal (T) and alternating field (AF) demagnetization. Solid and open circles represent horizontal (east–west and north–south) and vertical (upward–downward and north–south) components of magnetization, respectively. Numerals show temperatures of thermal demagnetization in °C or peak intensity of alternating magnetic field in mT.

clusters of antipodal directions, i.e. ($D = -47.1°$, $I = -27.4°$) and ($D = 103.8°$, $I = 27.6°$). These two directions are regarded as of normal or reversed polarity according to whether the inclination is negative or positive, as mentioned already, while the declinations are deflected to the east or to the west, respectively. Thus, these palaeomagnetic directions must correctly represent the direction of the ancient geomagnetic field.

3.4.3 Stage 3 (late Miocene to Pliocene)

The declination of the mean direction ($D = -32.3°$, $I = -44.9°$) in this stage is also clearly deflected to the west but the value is smaller than that for stage 2. Five reliable palaeomagnetic directions from both volcanic and sedimentary rocks of the Ba Volcanic Group, the Medrausucu Group and the Nadi Sedimentary Group were used in the calculation of the mean value.

The mean direction from four sites of early Pliocene age (see Table 3) is comparable with a mean direction ($D = -22°$, $I = -37°$) which had been compiled by James & Falvey (1978) from their own data and Tarling's data (1967). The circle of 95 per cent confidence level of this study satisfactorily overlaps that of their studies.

The samples from site no. 82-1 were collected to identify the Gilbert–Gauss boundary, which occurred at the expected stratigraphic level. The samples 82-1-1 to 82-1-3 were collected at 33.6, 30.9 and 37.5 m, respectively (of siltstone only) above the purple marker in the standard sequence of the Suva Marl. A change from reverse to normal polarity was found between samples 82-1-1 and 82-1-2 (Table 2). The age of this level was estimated from an extrapolated regression line as 3.46 Ma (Rodda *et al.* 1985), being in reasonable agreement with current polarity time-scales (Cox & Dalrymple 1967; Mankinen & Dalrymple 1979; Harland *et al.* 1982; Berggren *et al.* 1985).

Characteristics of the palaeomagnetic results from late Eocene to Pliocene from Viti Levu are summarized as follows.

(1) The calculated mean palaeomagnetic direction for each stage represents the primary magnetization.

(2) The declination is largely deflected to the west in stage 2. On the other hand, the declination in stage 1 is deflected toward north-northwest. The changes in declination took place in Oligocene and also after the end of middle Miocene.

(3) The changes in inclination are not remarkable between these stages.

4 ROTATION OF VITI LEVU

The mean virtual geomagnetic pole for each stage and its 95 per cent confidence level is shown in Table 3 and Fig. 5. Several studies have been carried out to estimate the Australian APWP (e.g. McElhinny, Embleton & Wellman 1974; Irving & Irving 1982; Idnurm 1985). The APWP from Australia by McElhinny *et al.* is most suitable for further discussion in this study, because the density of virtual geomagetic pole positions is higher for the late Eocene to Pliocene time than the other authors' and the palaeomagnetic sampling sites by McElhinny *et al.* are mostly in eastern Australia. However, it does not seem that there is much fundamental difference between the results of McElhinny *et al.* and those of Irving & Irving (1982) and Idnurm (1985) with reference to the present discussion. The APWP for the Pacific Plate by Suarez & Molnar (1980) is used in further discussion in the present study.

The palaeomagnetic direction change is caused both by tectonic movement and by apparent polar wander. If no tectonic movements in and around Viti Levu have taken place with respect to the Pacific Plate or the Australian Plate, the APWP from Viti Levu should agree precisely with that from either the Pacific Plate or the Australian Plate. Comparing the APWP from Viti Levu with that from the Australian Plate by McElhinny *et al.* or the Pacific Plate by

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Table 2. Palaeomagnetic results for Viti Levu.

Site	N	demag.	D (°)	I (°)	M (10^{-2}A/m)	α_{95} (°)	k	κ (10^{-4})	MDF (mT)	Qr	G.S. (μm)	rock
82-1-1	6	5mT	18.7	-54.2		19.4	15.6					
-2	6	5mT	111.2	30.5		12.0	31.9					
-3	5	5mT	19.5	-54.6		12.3	40.0					
82-1*(1)	17	5mT	-32.5	-52.0	1.9	12.6	9.0		3.8			siltstone
82-2	7	15mT	-64.5	-4.7	194.0	5.4	125.9	90.9	27.2	6.9	15-20	basalt
82-3	19	0	148.9	10.7	0.2	3.0	126.1	0.0	2.3	17.6		limestone
82-4	1	0	-162.2	23.5	238.0			3.1	9.3	1.9	8-10	basalt
82-5	14	0	-38.0	-4.4	130.4	3.7	116.4	68.2	2.0	6.0	50-100	basalt
82-6	5	6mT	1.0	-59.2	21.9	9.8	61.9	6.7	32.0	9.3	20-50	volcanics
82-7	9	5mT	-41.7	-26.2	185.0	2.4	461.2	219.2	2.2	3.0		sandstone
82-8	15	3mT	-1.4	-3.8	0.1	8.9	19.4	0.0	15.5	6.7		limestone
82-9	10	3mT	28.3	-6.5	31.5	13.7	13.4	3.5	26.0	3.9	20-30	volcanics
82-10	8	7.5mT	123.5	46.4	60.3	15.3	14.1	201.7	4.5	1.0	50-100	basalt
82-11	8	0	165.4	54.2	86.8	13.9	16.8	250.3	1.7	0.6	8-12	basalt
82-12	16	0	-16.0	-39.6	879.3	7.5	25.2	135.0	28.0	20.2	5-20	basalt
82-13	6	0	24.1	51.8	1003.7	18.4	14.2	236.0	6.5	14.6	10-30	volcanics
82-14	7	5mT	8.7	-31.8	137.1	3.5	298.4	201.4	4.8	1.8	50-100	micromonzonite
82-15	7	10mT	65.8	39.6	156.4	11.4	29.0	58.7	8.0	6.6	4-10	basalt
82-16	12	0	140.7	-8.2	28.2	62.9	1.4	28.9	23.0	1.1	2-4	gabbro
82-17	7	9mT	-1.2	-16.1	32.7	16.3	14.7	114.1	5.0	1.2	10-20	dacite
82-18	7	6mT	25.6	-30.9	33.5	8.5	51.4	74.8	12.5	1.0	15-25	andesite
82-19	11	6mT	-22.7	-36.9	48.1	6.0	58.9	122.5	3.0	1.1	40-60	andesite
85-1	8	300°C	-39.5	-9.3	1.1	18.9	9.5	14.0	10.0	1.4	5-50	mudstone
85-2	6	300°C	144.4	58.0	23.4	10.2	44.4		4.3		20-150	dolerite
85-3	10	550°C	40.0	80.9	0.1	18.6	7.7	3.4	44.0	4.7	5-20	sedimentary
85-4	11	500°C	143.7	49.0	1.5	14.9	10.3	111.3	12.0	0.3	10-100	sandstone
85-5	8	500°C	71.8	40.2	36.3	8.2	46.5		22.0		10-250	gabbro
85-6	7	10mT	20.0	-67.7	7.6	16.7	14.0	3.5	20.0	13.4	20-100	volcanics
85-7	4	15mT	-9.2	-61.0	11.2	9.7	90.3	65.4	12.0	1.6	5-100	mudstone
85-8	6	530°C	92.7	7.4	7.3	14.8	21.5	90.4	16.0	2.0	5-20	dacite
85-9	9	15mT	5.5	-33.0	8.7	3.9	173.6	93.4	7.5	1.2	20-200	sandstone
85-10	10	0	13.3	2.8	31.1	22.0	5.8	53.9	18.0	1.9	5-200	andesite
85-11	12	10mT	8.1	11.7	11.1	23.9	4.2	149.0	4.0	0.5	10-80	basalt
85-12	12	0	42.3	-10.8	53.2	30.0	3.0	211.5	4.0	0.8	10-40	sandstone
85-13	6	500°C	114.5	65.0	0.1	18.6	13.9	73.8	8.0	0.2	5-150	sandstone
85-14	11	20mT	116.3	31.8	0.3	17.9	7.5	140.3	2.7	0.7	4-40	andesite
85-15	13	15mT	61.6	-11.9	9.2	3.0	193.5	88.1	16.5	0.6	8-100	tuff
85-16	13	0	15.9	0.5	1.8	39.5	2.1	6.9	10.0	0.9		tuff
85-17	18	5mT	19.7	8.4	13.7	30.0	2.3	230.4	2.6	1.2	5-250	andesite
85-18	12	0	21.6	-16.3	3.7	29.3	3.2	20.2	6.6	0.6	10-100	sandstone
85-19	12	0	-13.9	-31.0	8.4	27.1	3.5	22.4	3.8	1.2	20-100	sandstone
85-20	11	30mT	-60.3	45.5	1.4	23.5	4.7	38.8	7.2	0.9	10-100	mudstone
85-21	11	0	6.8	-30.3	0.0	17.0	7.7	0.0	17.5	1.0	2-10	limestone

N: number of specimens, demag.: peak field of optimum demagnetization, *D*: declination of remanent magnetization, *I*: inclination of remanent magnetization, *M*: intensity of remanent magnetization, α_{95}: semiangle of cone of 95 per cent confidence, *k*: precision parameter, κ: volume susceptibility, MDF: median destructive field, *Qr*: Koenigsberger's ratio, G.S.: grain size. *(1) Both stable normal and reversed polarity samples in the site no. 82–1 are included. The reversed directions were inverted just in the opposite directions as the normal in calculation of the mean direction.

Suarez & Molnar, no agreement was found as shown in Fig. 5. This indicates that relative tectonic movement of Viti Levu has taken place with respect to both the Pacific Plate and the Australian Plate.

Expected declinations and inclinations for a representative point (178.0°E, 17.75°S) of Viti Levu, were calculated for several ages from the above-mentioned polar wander paths and were plotted in Fig. 6. The amount of tectonic movement of Viti Levu (rotation and movement parallel to the meridian), with respect to the Pacific Plate or the Australian Plate can be estimated from this figure.

Considerable disagreement between the palaeomagnetic and calculated declinations of Viti Levu indicates that the island of Viti Levu rotated as a micro-plate. Moreover, two rotational movement phases are detected from the changes in declination. The poles of these rotations are situated within or not far from Fiji, because the palaeomagnetic inclination of Viti Levu is very similar to both calculated inclinations from the Australian Plate and from the Pacific Plate as shown in Fig. 6. This fact also means that tectonic movement parallel to the meridian is hardly detected as the movement relative to either the Australian Plate or the Pacific Plate.

The anticlockwise rotation of about 75° of Viti Levu with respect to the Australian Plate since late Miocene is estimated from the palaeomagnetic declination of Viti Levu in stage 2 at −64° and from the expected declination according to the Australian APWP from 15 to 20 Ma at +10°. The intermediate value of difference in declination from late Miocene to Pliocene supports the existence of this anticlockwise rotation of Viti Levu. A difference of 35° was obtained between the palaeomagnetic declination of Viti Levu ($D = -32°$) in stage 3 and the expected declination of Viti Levu according to the Australian APWP ($D = +3°$) from 0 to 5 Ma. The palaeomagnetic results in stage 3 may represent a characteristic direction acquired during the period of rotation of Viti Levu.

The clockwise rotation of Viti Levu in Oligocene is estimated in the same manner. The declination of Viti Levu expected from the Australian Plate during this period is 20° estimated on the bases of interpolation between a 22° value for 25–35 Ma and a 19° value for 40–50 Ma. The value of

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Table 3. Palaeomagnetic directions for each age.

	D (°)	I (°)	α_{95} (°)	P	Site	V. G. P. Long. (°)	Lat. (°)	α_{95} (°)
Stage 1 (Late Eocene - Early Oligocene)								
	-1.4	-3.8	8.9	N	82-8	172.9	74.1	
	22.2	-44.8	13.7	N	82-9	65.3	67.8	
	-47.3	-60.6	18.9	N	85-1	47.1	43.3	
	-35.6	-58.0	10.2	N	85-2	46.6	52.7	
	169.4	56.6	18.6	R	85-3	21.6	68.5	
mean	-10.1	-47.9	29.4	(Ns = 5)		35.1	72.1	28.2
Stage 2 (Early - Middle Miocene)								
	-64.5	-4.7	5.4	N	82-2	93.8	25.0	
	161.3	10.6	3.0	R	82-3	119.9	68.0	
	-33.0	-2.1	3.7	N	82-5	111.6	53.6	
	-67.9	-37.1	9.8	N	82-6	73.2	26.3	
	65.8	39.6	11.4	R	82-15	58.4	-14.1	
	102.5	11.1	14.9	R	85-4	86.4	13.6	
	71.8	40.2	8.2	R	85-5	60.4	-8.9	
	-30.3	-50.8	9.5	N	85-6	55.6	59.4	
	-36.0	-37.3	9.7	N	85-7	76.9	56.0	
	92.7	7.4	14.8	R	85-8	85.3	3.7	
	121.9	43.5	18.6	R	85-13	69.1	35.9	
	106.3	19.4	17.9	R	85-14	83.2	18.5	
mean	-63.7	-28.3	18.1	(Ns ◘ 12)		79.0	29.2	17.6
Stage 3-1 (Late Miocene)								
	-39.2	-26.9	2.4	N	82-7	87.6	52.3	
Stage 3-2 (Early Pliocene)								
	-32.5	-52.0	12.6	N	82-1	54.8	57.3	
	123.5	46.4	15.3	R	82-10	66.4	37.4	
	165.4	54.2	13.9	R	82-11	32.7	68.7	
	-16.0	-39.6	7.5	N	82-12	67.9	74.3	
mean	-29.8	-49.3	16.7	(Ns = 4)				
Stage 3 (Late Miocene - Pliocene)								
mean	-32.3	-44.9	12.5	(Ns = 5)		65.5	59.2	16.8

Data are after tectonic correction. *D*: declination of remanent magnetization, *I*: inclination of remanent magnetization, α_{95}: semi-angle of cone of 95 per cent confidence, *P*: polarity of remanent magnetization, Ns: number of site for calculation of mean direction *in each stage*, V.G.P.: *virtual geomagnetic pole*, Long.: longitude of the position of V.G.P., Lat.: latitude of the position of V.G.P.

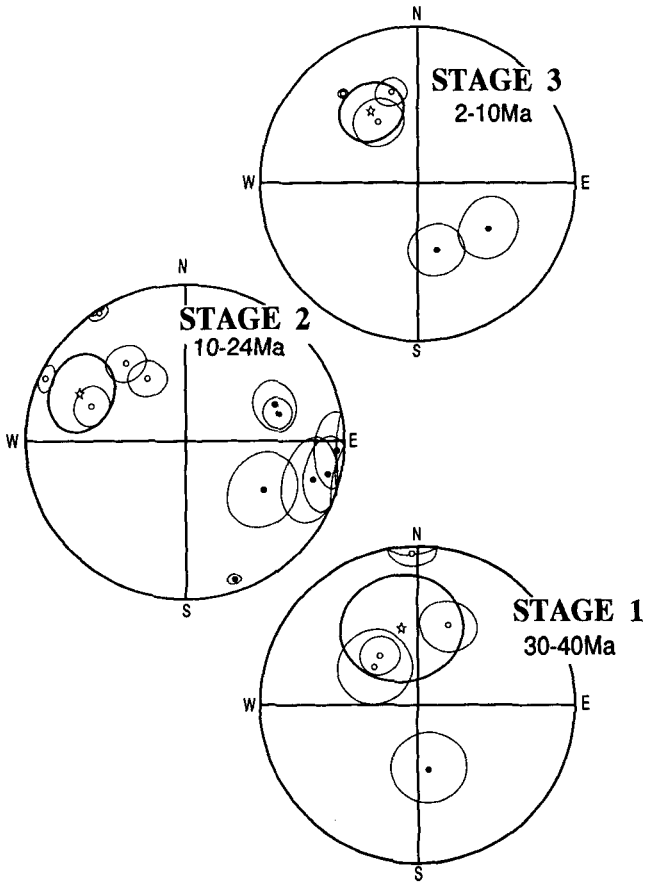


Figure 4. A summary of characteristic directions of magnetization *in each stage in equal area projection*. Site-mean directions and 95 per cent confidence circles are represented by dots and thin line, respectively. The mean directions in each stage and 95 per cent confidence circles are shown by a star and a thick line, respectively. Positive inclinations are shown as black dots and negative inclinations as open dots. Stage 1: late Eocene–early Oligocene, state 2: early–middle Miocene, and stage 3: late Miocene–Pliocene.

−30° was obtained by subtracting this mean value from the palaeomagnetic declination (−10°) of Viti Levu in stage 1. This value differs far greatly from the −75° declination difference estimated for stage 2 and suggests that a clockwise rotation of 45° took place during the Oliogcene (37–24 Ma).

5 TECTONIC IMPLICATIONS

As mentioned in the preceding section, the following conclusions can be drawn from the palaeomagnetic results of late Eocene to Pliocene in Viti Levu: (1) a clockwise rotation of 45° of Viti Levu occurred during Oligocene; (2) an anticlockwise rotation of 75° of Viti Levu has occurred since late Miocene; and (3) the Euler poles of these rotations are situated in or near Fiji. These rotations must closely relate to the origin and development of the complex physiographical and geological structure in the Fiji region. Timing, amount and direction of these rotations must be important as a clue to the origin and driving forces of these rotations.

A tectonic history from late Eocene to present in the Fiji region will be proposed which possibly explains these rotations. The Australian Plate is chosen as a reference frame in this palaeogeographic reconstruction (Fig. 7). Relative plate motion between the Pacific and the Australian Plates was computed from the rotaton poles and the angular velocities of the plates determined by Gordon & Jurdy (1986). Calculated relative motion between the two plates since 43 Ma at the position of Fiji can be represented by a convergent vector (8.6–8.8 cm yr^{-1}) directed almost perpendicular to the meridian in the hotspot reference frame. Reliability of this motion is also confirmed by the fact that both the trend of the inclination change with respect to the age and the absolute values of inclination at the reference point in Viti Levu calculated from the Australian virtual geomagnetic poles are very similar to those from the Pacific plate. The agreement of palaeomagnetic inclinations implies an agreement of palaeolatitudes. For consistency, the ages of all magnetic anomalies and the chronostratigraphic scale are reassigned to conform with the time-scale by Berggren *et al.* (1985), and are rounded off to the nearest million years. For the following discussion, information on the volcanism is mainly from to Kroenke (1984) and Rodda & Kroenke (1984) and the magnetic lineation patterns are from Malahoff *et al.* (1982a).

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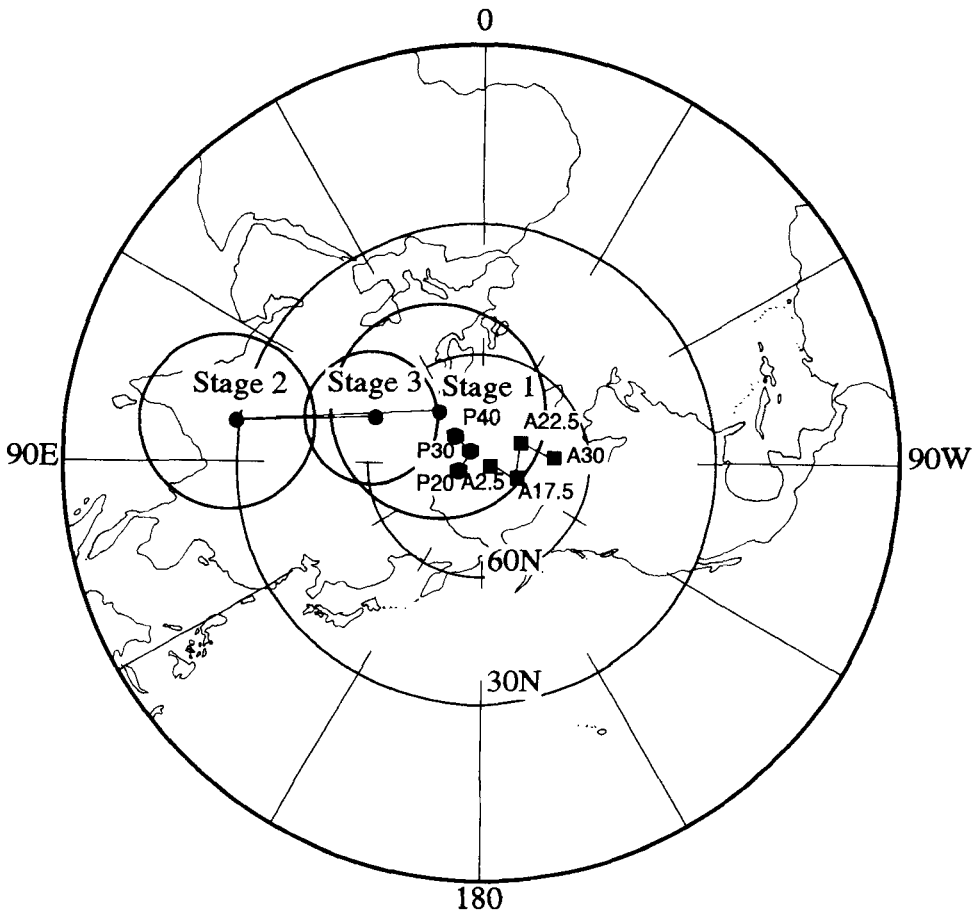


Figure 5. Virtual geomagnetic poles of Viti Levu from late Eocene to Pliocene. The mean pole positions for each stage (stage 1: late Eocene–early Oligocene, stage 2: early–middle Miocene, and stage 3: late Miocene–Pliocene) are shown by dots with a circle of α_{95}. Apparent polar wander path with ages in millions of years for the Pacific Plate (Suarez & Molnar 1980) and the Australian Plate (McElhinny *et al.* 1974) is expressed by hexagons and squares, respectively.

5.1 Early history (prior to 40 Ma)

The structural relationship between New Caledonia—the Norfolk Ridge and the Australian Continent had been fundamentally formed before 43 Ma (Fig. 1). Sea-floor spreading in the eastern margin of the Australian Continent began in late Cretaceous, displacing part of the eastern margin of the Australian Continent formed in the period from late Palaeozoic to early Mesozoic (Kroenke 1984). Formation of the New Caledonia Basin, displacing New Caledonia and the Norfolk Ridge to the northeast, seems to have been completed by early Palaeocene (Kroenke 1984). Formation of the Tasman Basin, displacing the Lord Howe Rise to the east continued until early Eocene (Weissel & Hays 1977). Direction of the plate motion for the Pacific Plate changed about 43 Ma ago (Morgan 1972; Clague & Jarrard 1973; Gordon & Jurdy 1986).

5.2 Late Eocene–early Oligocene (40–30 Ma)

Island-arc tectonism along the Vitiaz Ridge began in the period from late Eocene to early Oligocene and was probably caused by a new phase of subduction zone along the Vitiaz Trench with underthrusting of the Pacific Plate beneath the Australian Plate (Kroenke 1984). The initiation of this new subduction zone may be related to reorganization of the plate boundary between the Australian Plate and the Pacific Plate resulting from the change of motion of the Pacific Plate. Volcanism began along the Vitiaz Arc in the period from late Eocene to early Oligocene and continued until early Miocene. On Viti Levu, Vitiaz Arc magmatic activity is represented by a tonalite intrusion in the Yavuna Group (Hathway 1992).

5.3 Oligocene (37–24 Ma)

The timing (37–24 Ma) and the amount (45°) of clockwise rotation of Viti Levu give a clue to the origin of the rotation (Figs 7a and b). The South Fiji Basin formed during Oligocene by back-arc spreading marginal to the Lau–Tonga arc-trench system (Kamp 1986). Magnetic anomaly lineations 12 to 7A have been widely mapped (Watts *et al.* 1977; Davey 1982; Malahoff *et al.* 1982a) and anomaly 13 has been locally identified (Malahoff *et al.* 1982a), indicating that the sea-floor produced between 36 and 26 Ma involved a R–R–R triple junction. The clockwise rotation of Viti Levu with its northwest limb took place and the fan-shaped

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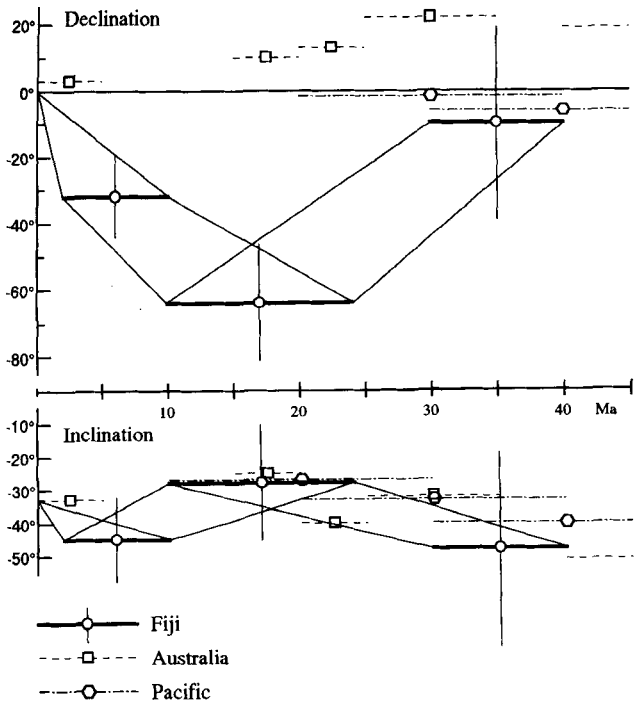


Figure 6. Change in declination and inclination of palaeomagnetic direction of Fiji as a function of the age. Mean palaeomagnetic directions in each stage are shown by circles. The declination and inclination error bars corresponding to the α_{95} are shown by thin lines. Calculated declination and inclination both from the Australian Plate (McElhinny *et al.* 1974) and from the Pacific Plate (Suarez & Molnar 1980) are represented by squares and hexagons, respectively. Time intervals in averaging data of Viti Levu, Australia and Pacific are shown by age bars.

northern half of the South Fiji Basin was formed with the progress of spreading the South Fiji Basin from a system involving a triple junction (Fig. 7b). The angle between the directions of anomaly 12 in the central part of eastern limb and in the central part of the northwestern limb is approximately 45°. This angle agrees with the amount of the rotation of Viti Levu. Clockwise rotation of Viti Levu and formation of the South Fiji Basin can be explained by introducing more plate boundaries, e.g. spreading ridges, between the Australian Plate and the Pacific Plate margins.

5.4 Early–middle Miocene (25–10 Ma)

Reorganization of the boundary between the Australian Plate and the Pacific Plate was completed in the early Miocene (Kroenke 1984). A brief period of volcanic activity and the deposition of coarse, clastic sediment during early to middle Miocene in the Vitiaz Arc and Viti Levu may be related to reactivation of a southwest-dipping subduction zone beneath the Vitiaz–Fiji Arc (Kroenke 1984; Rodda & Kroenke 1984).

Passive arc volcanism in the period from early to middle Miocene in the Vitiaz–Fiji Arc seems to be influenced by oblique subduction into and/or collision of the Ontong–Java Plateau with the Vitiaz Trench. This possibility is suggested from plate reconstruction using the data of rotation poles and angular velocities of the Australian Plate and the Pacific Plate by Gordon & Jurdy (1986). Oblique subduction is inferred from the evidence that ESE trend of the Vitiaz Trench is almost parallel to the relative plate motion and the collision is inferred from the palaeo-position of the Ontong–Java Plateau (Figs 7b and c).

The onset of volcanic activity along the combined Lau–Tonga volcanic axis occurred close to the Oligocene/Miocene time boundary and continued during Miocene (Kroenke 1984). Although convergence between the Australian Plate and the Pacific Plate was absorbed beneath the Lau–Tonga Arc throughout most of Miocene, the Three Kings Arc and the Loyalty Arc may have been the site of subduction from middle to late Miocene (Kroenke 1984). A middle to late Miocene age (15–9 Ma) for the volcanism along the Three Kings Ridge has been suggested on the basis of evidence from DSDP Site 206.

The missing western limb of the South Fiji Basin, magnetic anomaly sequence 7–13, was probably subducted under the Three Kings Ridge's complex during middle to late Miocene. Trench morphology as shown as a seismic-reflection profile taken across the South Loyalty Basin southwest of the Loyalty Ridge (Dubois *et al.* 1974) indicated northeast direction subduction. The subduction zones along the Loyalty Ridge and the Three Kings Ridges seem to have been joined by a transform fault forming the Cook Fracture Zone.

5.5 Late Miocene–Present (11–0 Ma)

A complex reorganization of plate boundaries occurred in the end of middle Miocene. The anticlockwise rotation of Viti Levu may be related to this reorganization. The collision of the Ontong–Java Plateau with the Vitiaz Trench may have stopped subduction along the Vitiaz Trench, close to the middle to late Miocene time boundary (Brocher 1985b). This collision probably led to a subsequent reversal of the subduction polarity to the southwest of the New Hebrides Arc to accommodate the ongoing convergence between the Australian and Pacific Plates (Brocher 1985b). The ongoing divergence of the currently active trenches was accompanied by back-arc spreading in the North Fiji Basin (Malahoff *et al.* 1982a). Palaeomagnetic data from the New Hebrides trench-arc system show an estimated clockwise rotation of 30° since late Miocene (Falvey 1978). N–S oriented magnetic anomalies 1–4 or 5 (0–8 or 10 Ma, respectively) are mapped in the southern portion of the North Fiji Basin (Malahoff *et al.* 1982a). Although the location, azimuth and rate of spreading of the ridges in the northern region of the basin are still uncertain (Brocher 1985b), the North Fiji Basin may have been formed since late Miocene involving a R–R–F triple junction (Figs 7c and d). The existence of an E–W oriented transform fault, the Fiji Transform Fault, to the north of Fiji was suggested on the bases of bathymetric (Green & Cullen 1973; Yaskawa *et al.* 1984) and seismotectonic (Eguchi 1984) data. Both the Fiji Transform Fault and the NE-trending Hunter Fracture Zone to the south of Fiji were identified as left-lateral shearzones based on focal mechanism solutions of shallow earthquakes (Eguchi 1984). The 75° anticlockwise rotation of Viti Levu since late Miocene time can be explained as a 'ball-bearing' rotation accommodated by antithetical transform faults, i.e. the Fiji Transform Fault and the Hunter

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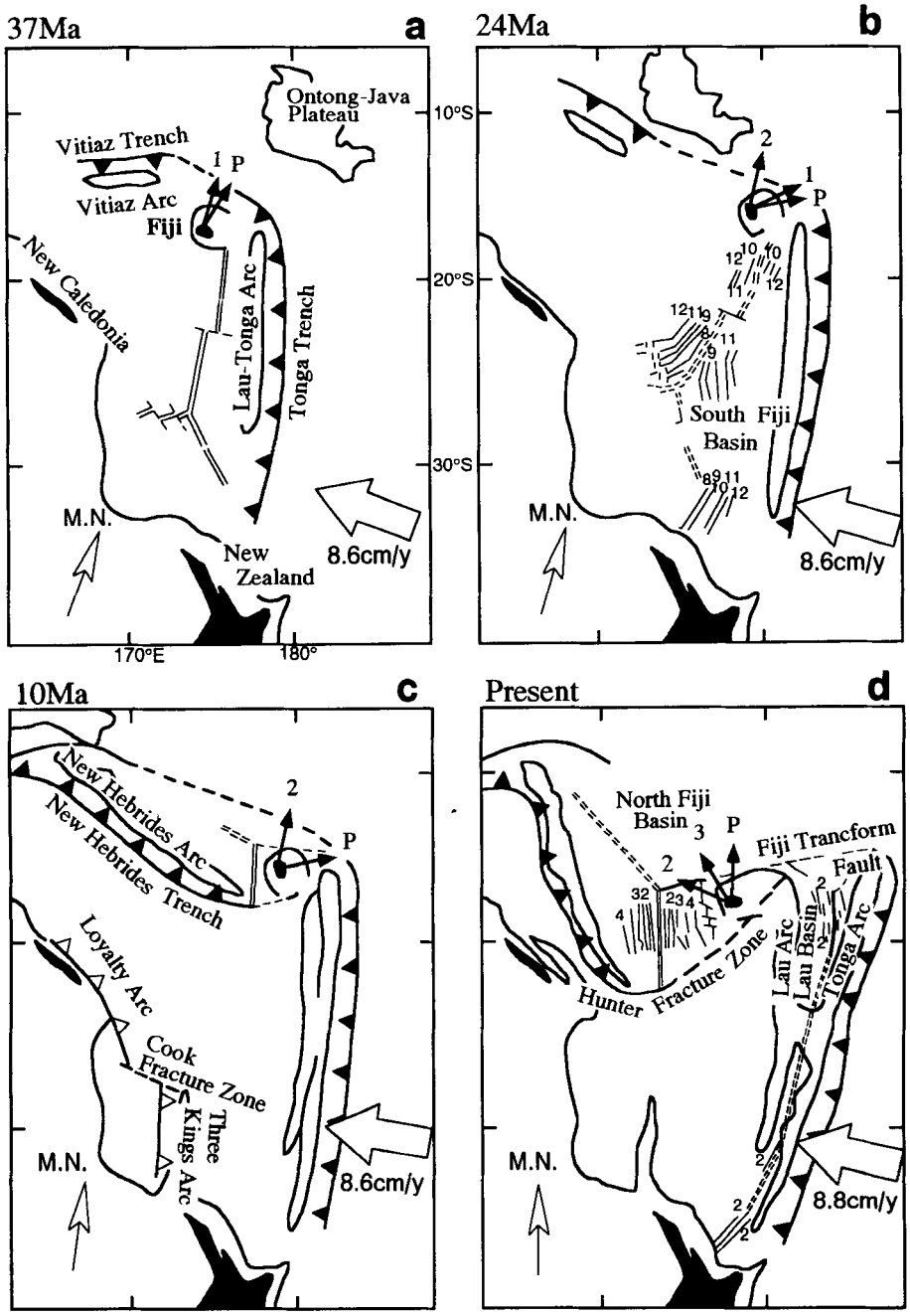


Figure 7. Schematic diagrams of tectonic evolution of the Fiji region in coordinate system fixed with respect to the Australian Plate. Arrows show obtained palaeomagnetic directions of Viti Levu for each stage (1: stage 1, late Eocene–early Oligocene, 2: stage 2, early–middle Miocene, 3: stage 3, late Miocene–Pliocene). Present north direction with respect to Viti Levu is represented by an arrow, P. Ticks show the location of the upper plate in a subduction zone. Failed subduction is shown with unfilled ticks. Sea-floor spreading centres are shown by double ruled lines, transform fault zones are by thick single-ruled lines, and by dashed lines where uncertain or inactive. Magnetic anomaly lineations are shown by thin lines with anomaly numbers taken from Malahoff *et al.* (1982a). The directions and rates of relative motion between the Australian Plate and the Pacific Plate are taken from Gordon & Jurdy (1986) and shown by unfilled arrows. The north direction according to the virtual geomagnetic north pole from McElhinny *et al.* (1974) is shown by an arrow in the west of the New Zealand.

Fracture Zone, as shown in Fig. 7(d). The possibility of such a mechanism was discussed for western North America (Beck 1976). The extent of this rotated rigid block is not certain; however, it may be suggested that one possible boundary is a transform fault zone situated to the west of Viti Levu (Brocher & Holmes 1985).

North trending magnetic lineations comprising anomalies 1–2 (0–3.5 Ma) are mapped over much of the Lau Basin. Although the tectonics of the northern portion of the Lau Basin is still uncertain, the initiation of the Lau–Tonga Trench and the ongoing divergence of currently active trenches have been accompanied by back-arc spreading in the Lau Basin since a time in late Miocene or early Pliocene (Weissel 1977; Malahoff *et al.* 1982a). Back-arc spreading

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has also been recognized in the Havre Trough (Malahoff *et al.* 1982a), south of the Lau Basin. A young age of spreading is supported by the seismotectonic study, by the geological study of the island and, moreover, by the following observations in the area; (1) the acoustic basement is shallow and rough, (2) the heat flow is high and variable, and (3) the sediment cover is thin (Weissel 1977).

Present plate boundaries are described by Brocher (1985b) as follows: (i) convergent boundaries include the Tonga–Kermadec Trenches, where the Pacific Plate is consumed, and the New Hebrides Trench, where the Australian Plate is consumed; (ii) divergent boundaries are north-trending sea-floor spreading in both the North Fiji Basin and the Lau Basin; and (iii) translational boundaries are the Fiji Transform Fault, which connects the spreading centre in the North Fiji Basin and the Tonga Trench, and the Hunter Fracture Zone which bounds the northern margin of the South Fiji Basin.

6 CONCLUSIONS

Reliable palaeomagnetic data obtained from 22 sites in Viti Levu can combine into three groups on the basis of geological evidence; i.e. ($D = -10°$, $I = -48°$), late Eocene–early Oligocene; ($D = -64°$, $I = -28°$), early–middle Miocene and ($D = -32°$, $I = -45°$), late Miocene–Pliocene. On the basis of the expected direction according to the Australian APWP, the tectonic movements of Viti Levu with respect to the Australian Plate was estimated from the remanent magnetization of the rocks. It was concluded that: (1) clockwise rotation of about 45° took place during Oligocene; (2) anticlockwise rotation of about 75° has occurred since the end of middle Miocene; and (3) these Euler poles are situated in or near Fiji.

To explain these rotations of Viti Levu a possible tectonic history of the Fiji region is proposed within the plate tectonic context. Clockwise rotation in Oligocene was closely related to the formation of the South Fiji Basin and anticlockwise rotation since the end of middle Miocene was closely related to the formation of the North Fiji Basin. The mechanisms of the two rotational movements seem to be different. The fan-shaped northern half of the South Fiji Basin was formed and clockwise rotation of Viti Levu took place with progressive spreading of the basin from a system involving the R–R–R triple junction. The anticlockwise rotation of Viti Levu is explained as 'ball-bearing' rotation accommodated by antithetical transform faults accompanied by the spreading of the North Fiji Basin.

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